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THE COMPLETE GUIDE

Regular Expressions

A Complete Guide to Regex for Developers

A complete, example-driven guide to regular expressions — from literals and character classes to quantifiers, groups, lookarounds, the Java API, practical patterns, and performance.

24	110+	12	6
Chapters	Examples	Diagrams	Modules

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Regular Expressions: The Complete Guide

First Edition · 2026

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MODULE 1 · REGEX FOUNDATIONS

CHAPTER 01

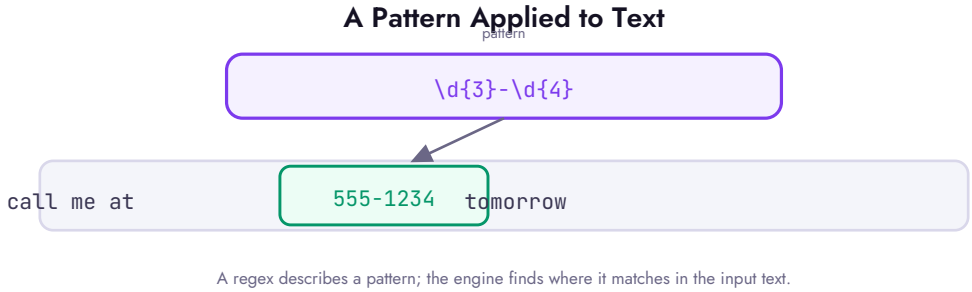
What Is a Regular Expression?

A compact language for describing text patterns, and where developers use it every day.

A **regular expression** — regex for short — is a small, dense language for describing patterns in text. Instead of writing loops to hunt for a phone number or validate an email, you describe the shape of what you want, and a regex engine finds it.

> **Regular expression**
> A sequence of characters that defines a search pattern. A regex engine matches the pattern against text to find, validate, extract, or replace — all from a compact declarative description.

A pattern in action



A regex describes a pattern; the engine finds where it matches in the input text.

The pattern `\d{3}-\d{4}` reads 'three digits, a dash, four digits'. Applied to a sentence, the engine locates `555-1234` without any manual character-by-character code.

Where you use regex

- **Validation** — is this a valid email, phone, or postcode?
- **Search** — find every date or URL in a document.
- **Extraction** — pull fields out of logs or text.
- **Replacement** — reformat or clean text in bulk.
- **Splitting** — break text on flexible delimiters.

Why learn it properly

Regex is famously easy to write badly — patterns that almost work, or that are impossible to read. Learned properly, it is one of the highest-leverage tools a developer has, turning pages of parsing code into a single expressive line.

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Describe the shape of text

A regex is a compact pattern language for matching, validating, extracting, and replacing text. It replaces hand-written parsing loops with a single declarative expression — powerful, but worth learning well.

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MODULE 1 · REGEX FOUNDATIONS

CHAPTER 02

Literals & the Match Process

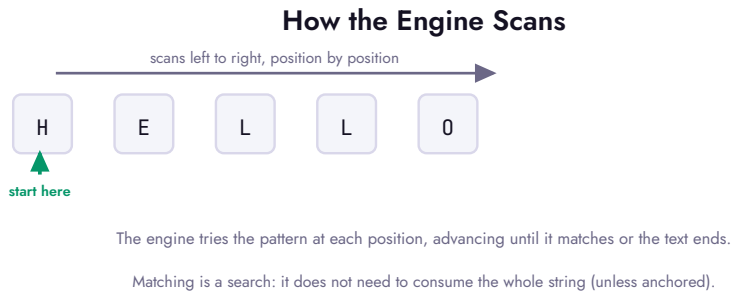
How the simplest patterns work, and the way the engine searches through text.

Before the special characters, start with the simplest regex of all: plain text. Understanding how the engine matches literals reveals how everything else works.

Literal characters

Most characters in a regex match themselves. The pattern `cat` matches the letters c, a, t in sequence — anywhere they appear, including inside `concatenate`. A regex is a **search**, not a whole-string comparison, unless you say otherwise.

How the engine scans



The engine tries the pattern at each position, advancing until it matches or the text ends.

The engine attempts the pattern at position 0. If it fails, it moves to position 1 and tries again, and so on. As soon as a match is found, it reports the location — it need not reach the end of the text.

Match vs search

Operation	Meaning
find / search	match anywhere in the text
matches (full)	the ENTIRE text must match

WATCH OUT

A common surprise: in Java, `matches()` requires the **whole** string to match the pattern, while `find()` looks for the pattern anywhere. Mixing them up is a frequent bug — know which you mean.

Case sensitivity

By default matching is case-sensitive: `cat` does not match `Cat`. A flag (covered later) makes matching case-insensitive when you need it.

Literals match, the engine searches

Ordinary characters match themselves, and the engine scans position by position until it finds a match. Distinguish searching anywhere (find) from requiring the whole string to match (matches).

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MODULE 1 · REGEX FOUNDATIONS

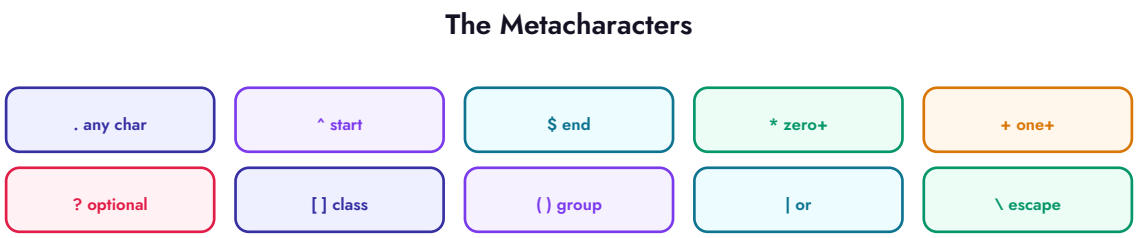
CHAPTER 03

The Metacharacters

The special characters that give regex its power — and why you must escape them.

The power of regex comes from **metacharacters** — characters that carry special meaning instead of matching literally. Learning which characters are special, and how to escape them, is the foundation of everything else.

The special characters



These characters have special meaning — to match one literally, escape it with a backslash.

These characters have special meaning; to match one literally, escape it with a backslash.

Metacharacter	Meaning
.	any single character
^ $	start / end anchors
* + ?	quantifiers (how many)
[ ]	character class
( )	grouping
|	alternation (OR)
\	escape the next character

Escaping

To match a metacharacter literally, precede it with a backslash. To match a literal dot, use `\.` — because a bare `.` means ‘any character’.

```regex
3.14      matches '3X14', '3.14', '3a14' (. = any char)
3\.14     matches only '3.14' (escaped dot = literal dot)
```

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WATCH OUT

The unescaped dot is the most common beginner mistake: `.` matches any character, not a literal period. Escape it as `\.` whenever you mean an actual dot.

The dot

The dot `.` matches any single character except a newline (by default). It is powerful but blunt — often a more specific class expresses intent better and matches more safely.

Special until escaped

Metacharacters like . ^ $ * + ? [ ] ( ) | carry special meaning. Escape any of them with a backslash to match it literally — especially the dot, which otherwise matches any character.

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MODULE 1 · REGEX FOUNDATIONS

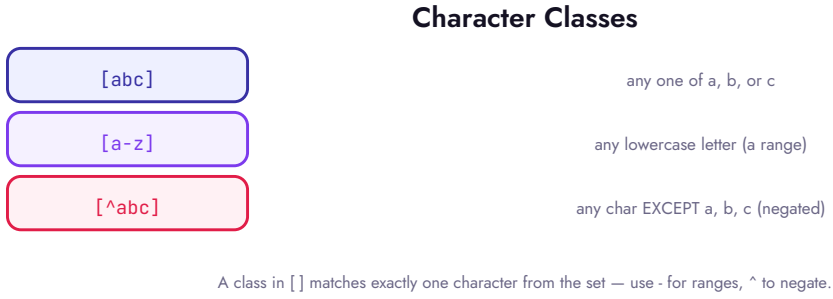
CHAPTER 04

Character Classes

Matching one character from a set, using brackets, ranges, and negation.

A **character class** — written in square brackets — matches exactly one character from a set you define. It is one of the most-used regex tools.

Sets, ranges, and negation



A class in brackets matches one character from the set — use a dash for ranges, a caret to negate.

Class	Matches
[abc]	a, b, or c
[a-z]	any lowercase letter
[A-Za-z0-9]	any letter or digit
[^0-9]	any character except a digit

Ranges

A dash inside a class defines a range: `[a-z]`, `[0-9]`, `[A-F]`. You can combine several ranges and individual characters in one class: `[a-zA-Z0-9_]`.

Negation

A caret as the first character negates the class: `[^aeiou]` matches any character that is **not** a vowel. Note the caret means 'start anchor' outside a class but 'negate' as the first character inside one.

Metacharacters inside classes

Inside a class, most metacharacters lose their special meaning — a dot in `[.]` is just a dot. Only a few (`]`, `\`, `^` at the start, and `-` between characters) need care.

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One character from a set

A character class in [ ] matches a single character from a set. Use dashes for ranges, a leading caret to negate, and remember most metacharacters are literal inside a class.

MODULE 2 · CORE BUILDING BLOCKS

CHAPTER 05

Predefined Character Classes

The shorthand classes for digits, word characters, and whitespace.

Some character sets are so common that regex provides shorthand for them. These **predefined classes** make patterns shorter and clearer.

The shorthands

Shorthand	Equivalent	Matches
\d	[0-9]	a digit
\w	[a-zA-Z0-9_]	a word character
\s	[\t\n\r]	whitespace
\D \W \S	negations	the opposite of each

Uppercase negates

The uppercase form is the negation of the lowercase: `\d` is a digit, `\D` is a non-digit; `\w` is a word character, `\W` is a non-word character. This pairing is consistent and easy to remember.

```regex
\d{4}          four digits (e.g. a year)
\w+            one or more word characters
\s*           any amount of whitespace (including none)
\d{3}-\d{4}   three digits, dash, four digits
```

A note on \w

By default `\w` covers ASCII letters, digits, and underscore — **not** accented or non-Latin letters. For full Unicode word characters you enable a Unicode flag (covered later), which matters for international text.

Shorthand for common sets
\d, \w, and \s match digits, word characters, and whitespace; their uppercase forms negate. They make patterns concise — but \w is ASCII-only unless you enable Unicode.

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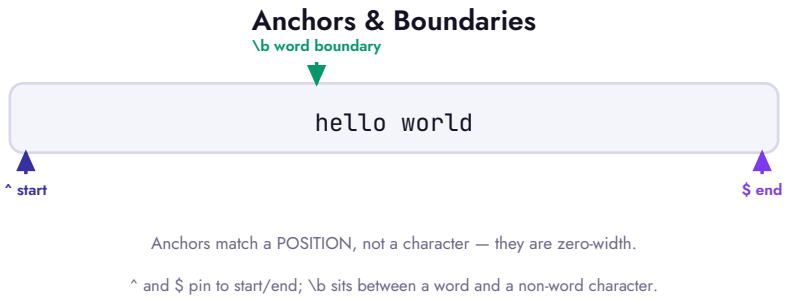
CHAPTER 06

Anchors & Boundaries

Matching positions rather than characters — start, end, and word boundaries.

Not everything in regex matches a character. **Anchors** and **boundaries** match a position — they are zero-width, asserting where in the text you are without consuming anything.

The positional assertions



Anchors match a position: ^ and $ pin to start/end, \b sits at a word boundary.

Anchor	Matches the position
^	at the start of the input (or line)
$	at the end of the input (or line)
\b	at a word boundary
\B	not at a word boundary

Start and end

^ and $ anchor a pattern to the start and end. `^\d+$` means 'the entire text is digits' — anchoring turns a loose search into a strict full-string check, which is essential for validation.

Word boundaries

`\b` matches the position between a word character and a non-word character. `\bcat\b` matches `cat` as a whole word but not inside `concatenate` — the fix for the classic 'matched a substring' problem.

```regex
\bcat\b     matches 'the cat sat', NOT 'concatenate'
^\d{6}$     the whole string is exactly six digits
```

Match where, not what
Anchors are zero-width: ^ and $ pin to start and end (turning a search into a full-string check), and \b asserts a word boundary so you match whole words, not substrings.

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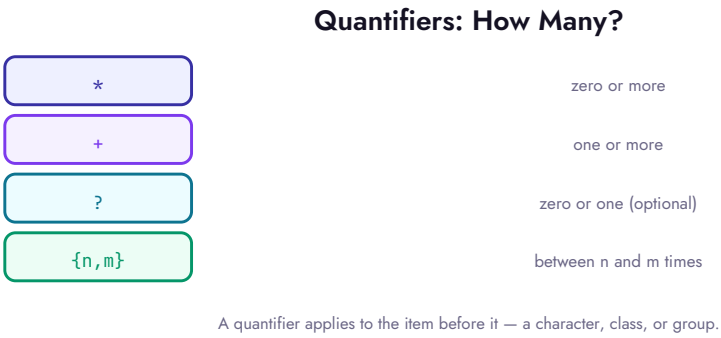
CHAPTER 07

Quantifiers

Specifying how many times something may repeat.

Quantifiers control repetition — how many times the preceding item may occur. They turn 'a digit' into 'one or more digits' and are what make regex concise.

The quantifiers



A quantifier applies to the item before it — a character, class, or group.

Quantifier	Repeats
*	zero or more times
+	one or more times
?	zero or one time (optional)
{n}	exactly n times
{n,}	n or more times
{n,m}	between n and m times

What they apply to

A quantifier applies to the single item immediately before it — a character, a class, or a group. In `ab+` the `+` repeats only `b`; to repeat `ab` you need a group: `(ab)+`.

regex

```
colou?r       matches 'color' and 'colour' (u optional)
\d{3,5}       three to five digits
[a-z]+        one or more lowercase letters
(ab)+         'ab', 'abab', 'ababab', ...
```

WATCH OUT

`*` and `?` allow **zero** occurrences, so a pattern of all-optional parts can match an empty string. When you require at least one, use `+` or `{1,}`.

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Control repetition

Quantifiers set how many times the preceding item repeats: * (0+), + (1+), ? (optional), and {n,m} for exact ranges. They apply to one item — group with ( ) to repeat more.

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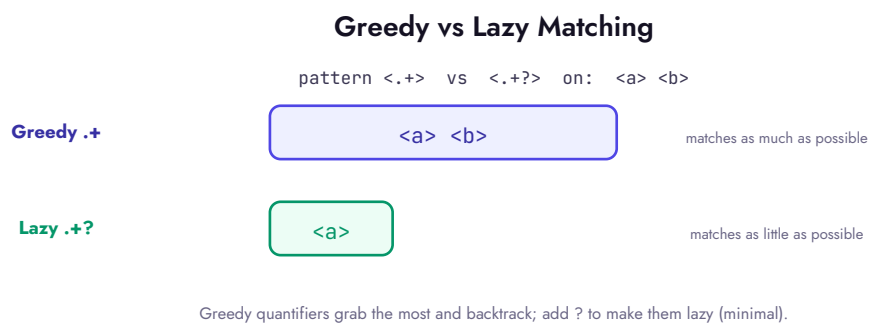
CHAPTER 08

Greedy, Lazy & Possessive

Why quantifiers grab as much as possible, and how to change that behaviour.

By default, quantifiers are **greedy** — they match as much as they can. This surprises people constantly, and understanding it (and the lazy and possessive alternatives) separates regex users from regex masters.

Greedy vs lazy



Greedy quantifiers grab the most and backtrack; add ? to make them lazy (minimal).

Given `<.+>` against `<a> <b>`, the greedy `.+` matches the whole `<a> <b>` — as much as possible — then backtracks to satisfy the final `>`. Adding `?` makes it lazy, matching just `<a>`.

Making a quantifier lazy

Append `?` to any quantifier to make it **lazy** (minimal): `*?`, `+?`, `??`, `{n,m}?`. Lazy quantifiers match as little as possible, expanding only when forced.

```regex
<.+>      greedy: matches '<a> <b>' in '<a> <b>'
<.+?>     lazy:   matches '<a>' then '<b>' separately
```

Possessive quantifiers

A **possessive** quantifier (append `+`: `*+`, `++`) matches greedily but **never backtracks**. It matches less often, but when it fits, it is faster and immune to certain performance disasters (covered later).

Type	Syntax	Backtracks?
Greedy	* + ?	yes
Lazy	*? +? ??	yes (expands)
Possessive	*+ ++ ?+	no

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Most, least, or none

Greedy quantifiers grab as much as possible and backtrack; adding ? makes them lazy (minimal); adding + makes them possessive (greedy, no backtracking). Choosing the right one fixes over-matching and slow patterns.

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MODULE 2 · CORE BUILDING BLOCKS

CHAPTER 09

Alternation & Grouping

Expressing choice with the OR operator, and using parentheses to control scope.

Two constructs let you build complex patterns from simple ones: **alternation** (a choice between options) and **grouping** (bundling a sub-pattern). Together they make regex composable.

Alternation with |



Parentheses group a sub-pattern; | means OR, and grouping controls what a quantifier applies to.

The pipe `|` means 'or': `cat|dog` matches either word. Alternation has low precedence, so `cat|dog` means the whole word `cat` or the whole word `dog`, not `ca`, then `t or d`, then `og`.

Grouping with parentheses

Parentheses bundle part of a pattern so a quantifier or alternation applies to the whole group. `(cat|dog)s` matches `cats` or `dogs`; `(ab)+` repeats `ab`.

regex

```
(cat|dog)s          'cats' or 'dogs'
(\d{1,3}\.){3}\d{1,3}   a dotted-quad like an IP address
gr(a|e)y            'gray' or 'grey'
```

Scoping alternation

WATCH OUT

Without grouping, alternation can span more than you intend. `^cat|dog$` means 'starts with cat' OR 'ends with dog' — use `^(cat|dog)$` to mean 'the whole string is cat or dog'.

Choice and composition

The pipe | expresses a choice between alternatives; parentheses group sub-patterns so quantifiers and alternation apply to the right scope. Together they compose simple patterns into complex ones.

MODULE 3 · GROUPS & CAPTURES

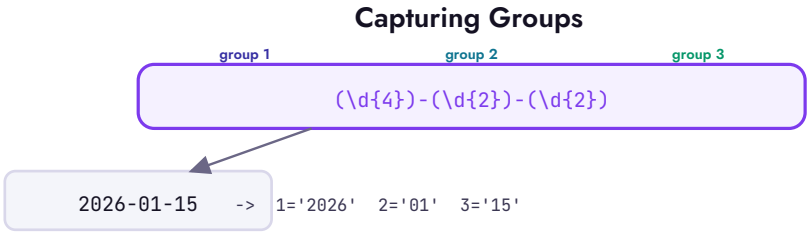
CHAPTER 10

Capturing Groups & Backreferences

Extracting matched sub-parts and referring back to them within the pattern.

Parentheses do more than group — they **capture**. Each captured group can be extracted after a match, or referred back to inside the same pattern. This is how you pull data out of text.

Numbered groups



Parentheses capture what they match into numbered groups you can extract or reference.

Parentheses capture what they match into numbered groups you can extract or reference.

Each opening parenthesis starts a new group, numbered left to right from 1. After matching, you retrieve each group's text — the way you extract the year, month, and day from a date.

```regex
(\d{4})-(\d{2})-(\d{2})

against  '2026-01-15'  captures:
  group 1 = 2026
  group 2 = 01
  group 3 = 15
```

Backreferences

A **backreference** matches the same text a group captured earlier, using `\1`, `\2`, and so on. This finds repetition: `(\w+) \1` matches a doubled word like `the the`.

```regex
(\w+)\s+\1     matches a repeated word: 'hello hello'
<(\w+)>.*</\1>  matches paired tags with the same name
```

Group zero

Group `0` is the entire match — useful when you want the whole matched text alongside the individual captures.

Capture to extract and reference

Parentheses capture matched text into numbered groups you can extract afterward, and backreferences (\1, \2) match the same text again within the pattern — ideal for finding repetition and paired structures.

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MODULE 3 · GROUPS & CAPTURES

CHAPTER 11

Named & Non-Capturing Groups

Readable named captures, and groups that organise without capturing.

Numbered groups work, but numbers get confusing as patterns grow. **Named groups** make captures readable, and **non-capturing groups** let you group without the overhead of a capture.

Named groups

Name a group with `(?<name>...)` and retrieve it by name instead of number — far clearer in code, and robust to reordering the pattern.

```regex
(?<year>\d{4})-(?<month>\d{2})-(?<day>\d{2})

// in Java: matcher.group("year")
// backreference by name: \k<year>
```

Non-capturing groups

When you need grouping only for scope — to apply a quantifier or alternation — but do not want to capture, use `(?:...)`. It groups without creating a numbered capture, keeping your group numbers clean and matching slightly faster.

```regex
(?:ab)+          groups 'ab' for the +, but captures nothing
(?:https?|ftp)://   group the scheme choice, no capture
```

Atomic groups

An **atomic group** `(?>...)` matches its content and then 'locks it in' — it will not backtrack into the group. Like possessive quantifiers, atomic groups are a tool for preventing the performance problems covered later.

TIP

Prefer `(?:...)` whenever you group only for scope. Fewer capturing groups means clearer intent and less work for the engine.

Name for clarity, (?:) for scope

Named groups (?<name>...) make captures self-documenting; non-capturing groups (?:...) organise a pattern without capturing; atomic groups (?>...) prevent backtracking. Use each to keep patterns clear and fast.

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MODULE 3 · GROUPS & CAPTURES

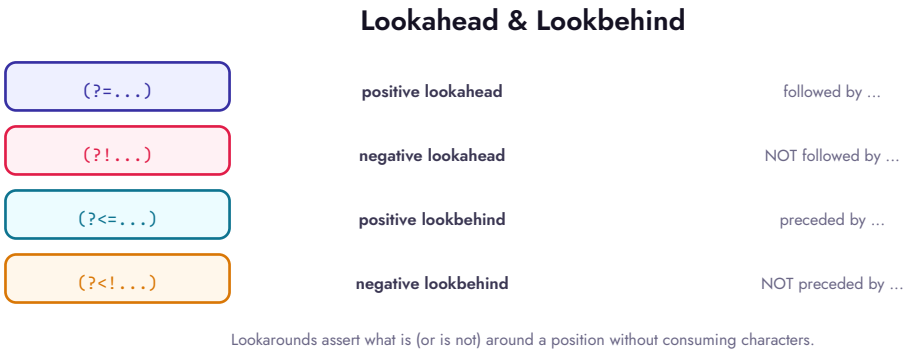
CHAPTER 12

Lookahead & Lookbehind

Zero-width assertions that check context without consuming characters.

Lookarounds are the most powerful — and most misunderstood — regex feature. They assert what is (or is not) around the current position without consuming any characters, letting you match based on context.

The four lookarounds



Lookarounds assert what is (or is not) around a position without consuming characters.

Syntax	Asserts the match is
(?=X)	followed by X (positive lookahead)
(?!X)	NOT followed by X (negative lookahead)
(?<=X)	preceded by X (positive lookbehind)
(?<!X)	NOT preceded by X (negative lookbehind)

Why zero-width matters

A lookaround checks a condition but does not include the checked text in the match. This lets you match a position defined by its surroundings — for example, a number followed by a currency symbol, without capturing the symbol.

```regex
\d+(?= dollars)     matches the number in '50 dollars' (not ' dollars')
(?<=\$)\d+          matches the number in '$50' (not the '$')
\b\w+\b(?!,)         a word not followed by a comma
```

A classic use: password rules

Lookaheads express ‘must contain’ rules compactly. A password requiring a digit and a letter can be checked with stacked lookaheads at the start, each asserting a condition over the whole string.

regex

```
^(?=.*\d)(?=.*[A-Za-z]).{8,}$
// at least one digit, one letter, and 8+ characters
```

Assert context, consume nothing

Lookaheads (?=)(?!) and lookbehinds (?<=)(?<!) assert what surrounds a position without consuming it. They match by context — powering currency matching, password rules, and precise boundaries.

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MODULE 4 · REGEX IN JAVA

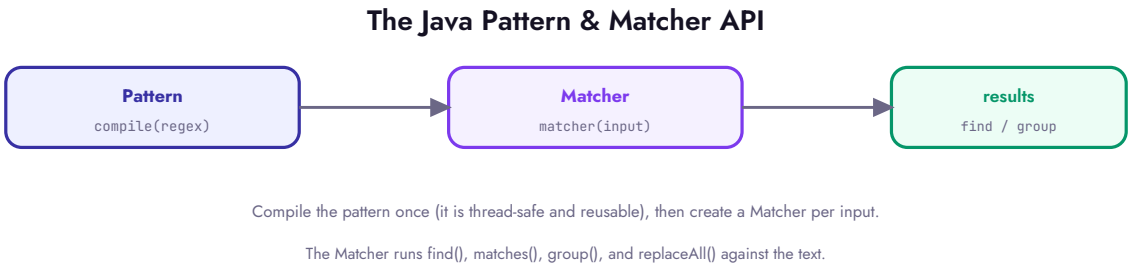
CHAPTER 13

The Pattern & Matcher API

Java's two core regex classes and the right way to use them.

Java's regex support lives in the `java.util.regex` package, built around two classes: `Pattern` (a compiled regex) and `Matcher` (the engine applied to an input).

Compile, then match



Compile the pattern once (reusable and thread-safe), then create a Matcher per input.

```java
Pattern pattern = Pattern.compile("\\d{3}-\\d{4}");
Matcher matcher = pattern.matcher("call 555-1234 now");

if (matcher.find()) {
    System.out.println(matcher.group());   // 555-1234
}
```

Compile once, reuse

Compiling a regex is relatively expensive, and a `Pattern` is immutable and **thread-safe**. Compile it once — ideally into a `static final` field — and reuse it, creating a lightweight `Matcher` per input.

```java
private static final Pattern PHONE =
    Pattern.compile("\\d{3}-\\d{4}");
// reuse PHONE.matcher(input) everywhere
```

WATCH OUT

Calling `Pattern.compile` inside a loop or hot method recompiles the regex every time — a real performance drain. Compile once and store the `Pattern`.

The convenience shortcut

`String` methods like `matches()`, `split()`, and `replaceAll()` accept regex directly. They are convenient for one-off use but recompile the pattern each call — fine occasionally, wasteful in a loop.

Pattern compiles, Matcher runs

Pattern.compile() builds a reusable, thread-safe compiled regex; a Matcher applies it to one input. Compile once into a constant and reuse it — never recompile in a loop.

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MODULE 4 · REGEX IN JAVA

CHAPTER 14

Finding, Matching & Replacing

The core Matcher operations and the crucial matches-versus-find distinction.

The `Matcher` offers several ways to apply a pattern. Choosing the right one — and understanding the difference between matching the whole string and finding within it — is essential.

The core methods

Method	Does
matches()	the WHOLE input must match
find()	find the next match anywhere
lookingAt()	match at the start (not whole)
group()	the matched text (or a group)
replaceAll()	replace every match

Iterating over matches

Call `find()` repeatedly to walk through every match in the input, extracting each with `group()` — the standard way to pull all occurrences of a pattern from text.

```java
Matcher m = Pattern.compile("\\d+").matcher("a1 b22 c333");
while (m.find()) {
    System.out.println(m.group());   // 1, then 22, then 333
}
```

Accessing groups and positions

Within a match, `group(n)` returns a captured group, and `start()/end()` give the match's positions — useful for highlighting or slicing the original text.

Replacing

```java
String masked = "card 1234-5678"
    .replaceAll("\\d", "*");   // 'card ****-****'
```

WATCH OUT

Remember: `matches()` anchors implicitly to the whole string, while `find()` searches. `"abc123".matches("\\d+")` is false — the whole string is not digits.

Match the whole or find within

matches() requires the entire input to match; find() locates matches anywhere and can iterate; group() extracts them and replaceAll() replaces them. The matches-vs-find distinction is the one to internalise.

MODULE 4 · REGEX IN JAVA

CHAPTER 15

Flags & Modifiers

Changing how a pattern matches — case, multiline, dotall, and comments.

Flags modify how the whole pattern matches, without changing the pattern itself. A few flags handle the most common needs — case-insensitivity, multiline text, and readability.

The common flags

Flag	Effect
CASE_INSENSITIVE	ignore case
MULTILINE	^ and $ match each line
DOTALL	. also matches newlines
COMMENTS	allow whitespace & comments in the pattern

Applying flags in Java

```java
Pattern p = Pattern.compile("error",
    Pattern.CASE_INSENSITIVE | Pattern.MULTILINE);
// combine flags with | (bitwise OR)
```

Inline flags

You can also set flags inside the pattern with `(?i)` for case-insensitive, `(?m)` for multiline, `(?s)` for dotall — useful when the pattern is a string you cannot pass flags with.

```regex
(?i)hello        matches 'hello', 'HELLO', 'Hello'
(?s).*           . now matches across newlines too
```

Multiline and dotall explained

- **MULTILINE** — ^ and $ match at every line break, not just the whole input's ends.
- **DOTALL** — the dot matches newlines too (by default it does not).
- **COMMENTS** — lets you space out a complex pattern and add # comments for readability.

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Modify how the pattern matches

Flags change matching behaviour globally: CASE_INSENSITIVE, MULTILINE (^/$ per line), DOTALL (. matches newlines), and COMMENTS (readable patterns). Set them via Pattern.compile or inline as (?i), (?m), (?s).

MODULE 4 · REGEX IN JAVA

CHAPTER 16

Splitting & Extracting Data

Using regex to break text apart and pull structured data out of it.

Two of regex's most practical jobs are **splitting** text on flexible delimiters and **extracting** structured fields. Both turn messy text into usable data.

Splitting

`String.split(regex)` breaks text wherever the pattern matches — far more flexible than splitting on a single character. You can split on 'any run of whitespace' or 'a comma with optional spaces'.

```java
"a,  b,c ,  d".split("\\s*,\\s*");
// ["a", "b", "c", "d"]  — handles the messy spacing

"one   two\tthree".split("\\s+");
// ["one", "two", "three"]  — any whitespace
```

Extracting with groups

To pull fields out of a structured line, match with capturing groups and read each group. This parses log lines, key-value pairs, and simple records without a full parser.

```java
Pattern p = Pattern.compile(
    "(?<level>\\w+)\\s+\\[(?<time>[^\\]]+)\\]\\s+(?<msg>.*)");
Matcher m = p.matcher("ERROR [10:32:01] disk full");
if (m.find()) {
    m.group("level");  // ERROR
    m.group("time");   // 10:32:01
    m.group("msg");    // disk full
}
```

Extracting all matches

Combine `find()` in a loop with groups to extract every record from a block of text — for example, every email address or every timestamp in a document.

WATCH OUT

`split` with a limit of 0 drops trailing empty strings, which can surprise you. Use a negative limit to keep them when empty fields are meaningful.

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Break apart and pull out

split(regex) divides text on flexible patterns, and capturing (or named) groups extract structured fields from each match. Together they parse logs, records, and messy input without a bespoke parser.

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MODULE 4 · REGEX IN JAVA

CHAPTER 17

Escaping in Java Strings

The double-backslash trap and how Java string literals interact with regex.

The single most confusing thing about regex in Java is **backslashes**. Because both Java strings and regex use the backslash as an escape character, they collide — and the result trips up nearly everyone at first.

Two layers of escaping

A regex `\d` must be written as `"\\d"` in a Java string. The Java compiler first processes the string, turning `\\` into a single backslash, and the regex engine then sees `\d`.

You want (regex)	Java string literal
\d	"\\d"
\.	"\\."
\\	"\\\\"
\s+	"\\s+"

WATCH OUT

Forgetting the doubling is the classic Java regex error. `"\d"` does not even compile as a Java string (invalid escape); you must write `"\\d"`.

Text blocks help

Java text blocks (triple-quoted strings) still require backslash doubling for regex, but improve readability for multi-line patterns. There is no way around the doubling in a normal string literal.

Escaping user input

When a user's text must be matched **literally** (not as a pattern), never concatenate it into a regex — escape it with `Pattern.quote()`, which neutralises any metacharacters it contains.

```java
String userText = "a.b (c)";
Pattern p = Pattern.compile(Pattern.quote(userText));
// matches the literal string, metacharacters and all
```

Double the backslashes

In Java, a regex backslash must be doubled in a string literal (\d becomes "\\d") because both layers escape. Use Pattern.quote() to match untrusted input literally rather than as a pattern.

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MODULE 5 · PRACTICAL PATTERNS

CHAPTER 18

Validating Input

Building anchored patterns to check whether input has the required form.

One of regex's most common jobs is **validation** — confirming that input has the right shape. The key technique is anchoring, so the entire input must conform.

Anchor for validation

Validation patterns must be anchored with `^` and `$` (or use `matches()`, which anchors implicitly). An unanchored pattern would pass any string that merely contains a valid part.

```java
// six-digit PIN code
"110001".matches("^\\d{6}$");     // true
"abc110001xyz".matches("\\d{6}"); // false (matches() is whole-string)
```

Common validation patterns

Validate	Pattern (simplified)
digits only	^\d+$
username	^[a-zA-Z0-9_]{3,20}$
hex colour	^#[0-9A-Fa-f]{6}$
Indian PIN	^[1-9]\d{5}$

The email caution

WATCH OUT

A fully correct email regex is enormous and still imperfect. For real email validation, use a simple sanity-check pattern and then **verify by sending a confirmation** — do not try to perfectly validate email with regex alone.

Keep validators readable

Complex validators become unmaintainable one-liners. Break them into pieces, add comments (with the COMMENTS flag), or combine simpler checks in code. A readable validator you can trust beats a clever one you cannot.

Anchor to validate the whole

Validation patterns must be anchored (^…$ or matches()) so the entire input conforms. Keep them readable, and for email prefer a light regex plus a confirmation step over a monstrous pattern.

MODULE 5 · PRACTICAL PATTERNS

CHAPTER 19

Extracting & Parsing Data

Turning semi-structured text — logs, feeds, documents — into structured data.

Beyond validation, regex excels at **extraction** — pulling structured pieces out of semi-structured text like log files, reports, and scraped content.

A parsing mindset

Identify the fixed parts (delimiters, labels) and the variable parts (the data you want), then capture the variable parts with groups. Named groups keep the result readable when there are several fields.

```java
// parse 'name=Ada; age=36; city=Delhi'
Pattern kv = Pattern.compile("(\\w+)=([^;]+)");
Matcher m = kv.matcher(input);
Map<String,String> fields = new HashMap<>();
while (m.find()) {
    fields.put(m.group(1), m.group(2).trim());
}
```

Extracting all occurrences

Loop with `find()` to collect every match — all URLs, all dates, all mentions — from a block of text. This is the workhorse pattern for log analysis and light scraping.

```regex
https?://[^\s]+          // URLs in text
\b\d{4}-\d{2}-\d{2}\b   // ISO dates
@\w+                     // @mentions
```

Know when to stop

WATCH OUT

Regex parses **regular** patterns, not nested structures. Do not parse HTML, deeply nested JSON, or full programming languages with regex — use a real parser. Regex is for flat, predictable text.

Capture the variable parts

Extraction pins the fixed structure and captures the variable data with (named) groups, looping with find() to collect every occurrence. Use it for flat, predictable text — not nested formats like HTML.

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MODULE 5 · PRACTICAL PATTERNS

CHAPTER 20

Search & Replace Techniques

Powerful substitutions using group references and computed replacements.

Regex replacement goes far beyond swapping one word for another. Using **group references** in the replacement, and computing replacements in code, you can reformat and transform text precisely.

Referencing groups in the replacement

In the replacement string, `$1`, `$2`, (or `${name}`) insert what those groups captured — letting you reorder and reformat matched text.

```java
// reformat 2026-01-15 -> 15/01/2026
"2026-01-15".replaceAll(
    "(\\d{4})-(\\d{2})-(\\d{2})", "$3/$2/$1");
```

Escaping in replacements

WATCH OUT

In the replacement string, `$` and `\` are special. If the replacement text comes from a user or data, escape it with `Matcher.quoteReplacement()` to avoid corrupting the output.

Computed replacements

When the replacement must be **calculated** per match (not a fixed template), use `Matcher.replaceAll` with a function, which receives each match and returns its replacement.

```java
// upper-case every word
String out = Pattern.compile("\\w+")
    .matcher(text)
    .replaceAll(mr -> mr.group().toUpperCase());
```

First vs all

`replaceFirst` replaces only the first match; `replaceAll` replaces every match. Choose deliberately — replacing all when you meant first (or vice versa) is a subtle bug.

Reference and compute replacements

Replacement strings can reference captured groups ($1, ${name}) to reformat text, and a replacement function computes each substitution. Escape untrusted replacements, and pick replaceFirst vs replaceAll deliberately.

MODULE 5 · PRACTICAL PATTERNS

CHAPTER 21

Unicode & Internationalization

Matching real-world international text correctly, beyond ASCII.

The default shorthands like `\w` and `\d` are ASCII-only — fine for English, but wrong for names, scripts, and text from the rest of the world. Regex has full Unicode support when you ask for it.

The ASCII default problem

WATCH OUT
By default `\w` excludes accented and non-Latin letters, so `[a-zA-Z]` and `\w` reject valid names like `Zoë` or text in Hindi and Tamil. Do not assume ASCII for user-facing input.

Unicode property classes

Match by Unicode **property** using `\p{...}` — for example `\p{L}` for any kind of letter in any script, `\p{N}` for any number, and `\p{Lu}` for uppercase letters.

```regex
\p{L}+        one or more letters, any language
\p{N}         any numeric character
\p{L}\p{M}*   a letter plus any combining marks
```

Making shorthands Unicode-aware

In Java, the `UNICODE_CHARACTER_CLASS` flag (or inline `(?U)`) makes `\w`, `\d`, and `\s` respect Unicode, so `\w` matches letters in any script rather than only ASCII.

```java
Pattern p = Pattern.compile("\\w+",
    Pattern.UNICODE_CHARACTER_CLASS);
```

Beware of surprises

Unicode has subtleties — combining characters, normalization forms, and characters that look alike but differ. For serious international handling, normalize input first and prefer property classes over hand-built ranges.

Ask for Unicode explicitly
Default \w and \d are ASCII-only and reject international text. Use \p{L}, \p{N} property classes or the UNICODE_CHARACTER_CLASS flag so patterns match names and scripts from every language.

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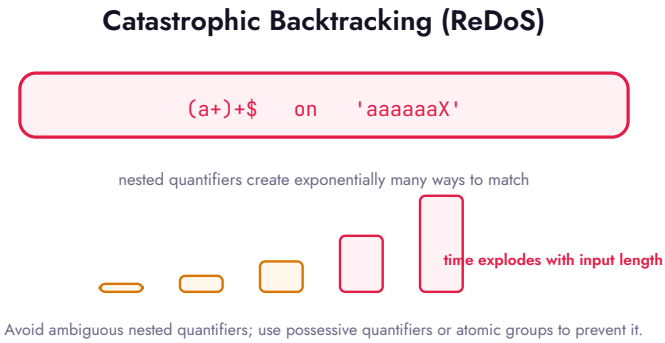
CHAPTER 22

Performance & Catastrophic Backtracking

Why some patterns hang on certain inputs, and how to write safe, fast regex.

Regex is usually fast, but a poorly written pattern can take exponential time on certain inputs — hanging your application or enabling a denial-of-service attack. This is **catastrophic backtracking**, and every developer must recognise it.

What goes wrong



Nested quantifiers create exponentially many ways to match; time explodes with input length.

When a pattern has **nested or overlapping quantifiers** — like `(a+)+` — and the input almost matches, the engine tries an exponential number of ways to split the text before giving up. A few dozen characters can take seconds or minutes.

WATCH OUT

Patterns like `(a+)+$`, `(.*)*`, or `(\d+)*` on non-matching input are the classic ReDoS (Regular expression Denial of Service) traps. Untrusted input plus such a pattern is a real vulnerability.

How to avoid it

- Avoid nesting quantifiers over overlapping alternatives.
- Make patterns **specific** — the less ambiguity, the fewer paths to try.
- Use **possessive quantifiers** (++) or **atomic groups** (`(?>...)`) to forbid backtracking.
- Anchor patterns so the engine cannot wander.
- Limit input length before matching untrusted text.

A safer rewrite

```regex
(a+)+$        DANGEROUS: catastrophic on 'aaaaaaaaX'
a+$           SAFE: same intent, no nested quantifier
"[^\"]*+"     possessive: fast, no backtracking
```

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Beware nested quantifiers

Nested or overlapping quantifiers can cause exponential (catastrophic) backtracking — a hang or ReDoS on crafted input. Keep patterns specific and anchored, use possessive/atomic constructs, and bound untrusted input length.

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MODULE 6 · MASTERY

CHAPTER 23

Debugging & Testing Regex

Building patterns incrementally and proving they work — and fail — correctly.

A regex that looks right often is not. Because patterns are dense and easy to get subtly wrong, disciplined building and testing matter more here than almost anywhere else.

Build incrementally

Do not write a complex pattern in one go. Start with the core, test it, then add anchors, quantifiers, and edge cases one at a time — checking after each addition. A regex grown piece by piece is far easier to get right.

Use a tester

Interactive regex testers (regex101 and similar) show matches live, explain each token, and highlight backtracking — invaluable for understanding what a pattern actually does versus what you intended.

Test both matches and non-matches

The dangerous case is not what your pattern matches, but what it **wrongly** matches or rejects. Always test strings that should fail, not just ones that should pass — that is where bugs hide.

```java
@Test
void pinValidation() {
    Pattern p = Pattern.compile("^[1-9]\\d{5}$");
    assertTrue(p.matcher("110001").matches());   // valid
    assertFalse(p.matcher("011000").matches());  // leading zero
    assertFalse(p.matcher("11000").matches());   // too short
    assertFalse(p.matcher("1100011").matches()); // too long
}
```

Comment complex patterns

For anything non-trivial, use the COMMENTS flag (or `(?x)`) to lay the pattern out over multiple lines with explanations — your future self will thank you.

Grow it, then break it

Build patterns incrementally with a live tester, and test the failure cases as rigorously as the success cases — wrong matches hide there. Comment complex patterns with the COMMENTS flag for maintainability.

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CHAPTER 24

Best Practices & Cheat Sheet

A consolidated set of habits and a quick reference to keep at hand.

You now know regex from literals to lookarounds. This closing chapter distils the habits that keep patterns correct, fast, and maintainable — plus a compact reference.

Best practices

- **Compile once** — store a `Pattern` in a constant; never recompile in a loop.
- **Anchor validators** — use `^...$` so the whole input conforms.
- **Be specific** — prefer precise classes over `.`; specificity is speed and safety.
- **Avoid nested quantifiers** — they invite catastrophic backtracking.
- **Prefer non-capturing** `(?:...)` when you group only for scope.
- **Comment complex patterns** and test failure cases.
- **Do not parse nested formats** (HTML, JSON) — use a real parser.

Quick reference

Construct	Meaning
. \d \w \s	any char / digit / word / space
[abc] [^abc] [a-z]	class / negated / range
^ $ \b	start / end / word boundary
* + ? {n,m}	quantifiers (greedy)
*? +?	lazy quantifiers
( ) (?:) (?<n>)	capture / non-capture / named
|	alternation (OR)
(?=) (?!) (?<=) (?<!)	look ahead / behind
\1 \k<n>	backreference by number / name

The mindset

Regex rewards clarity over cleverness. A readable pattern you and your team can maintain is worth more than a dazzling one-liner nobody dares touch. Build deliberately, test thoroughly, and reach for a real parser when the text stops being regular.

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Clear, specific, tested

Compile once, anchor validators, stay specific, avoid nested quantifiers, and never parse nested formats with regex. Keep the cheat sheet handy — and favour readable patterns over clever ones.

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